

Holographic Complexity

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2 Holographic Complexity and Extended Thermodynamics

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Introduction

The AdS/CFT correspondence

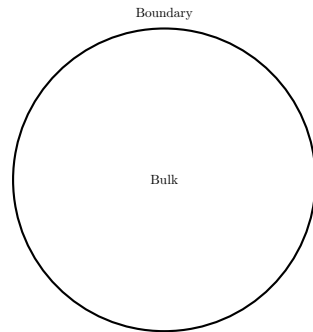


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- 'bulk' gravity theory in $d + 1$ dimensional asymptotically AdS spacetime $\leftrightarrow$ 'boundary' d dimensional conformal field theory

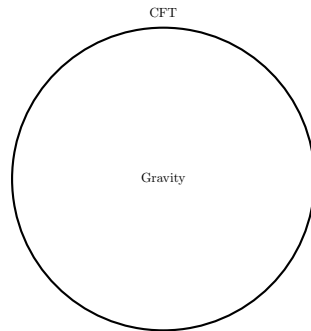


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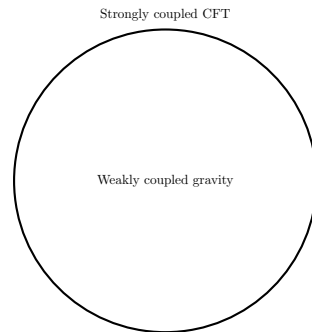


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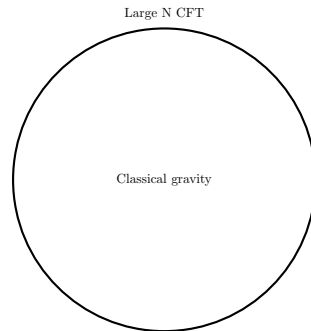


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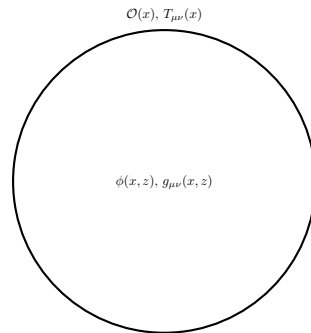


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- boundary thermal state $\leftrightarrow$ black hole (above Hawking-Page transition)

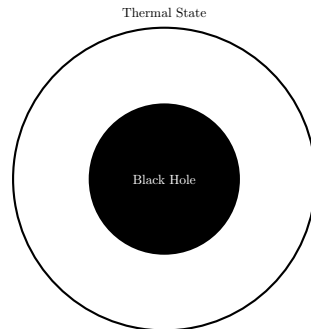


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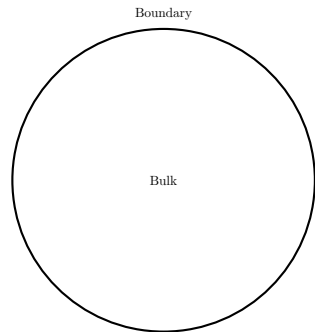


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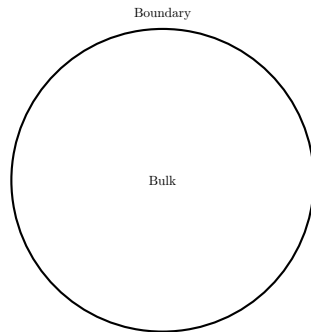


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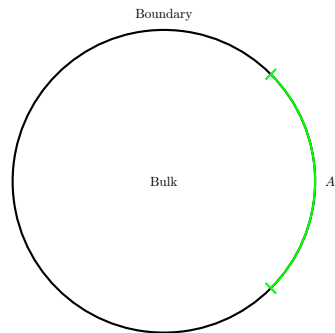


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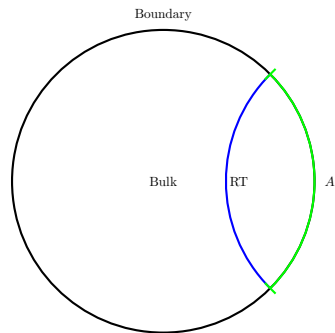


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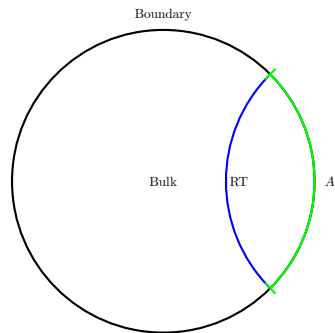


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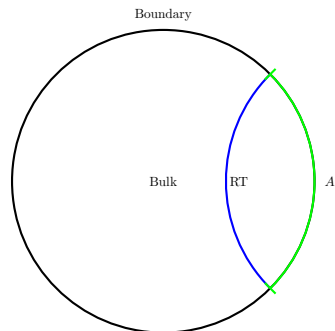


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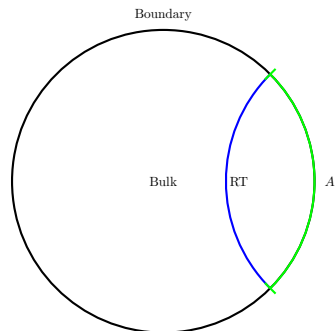


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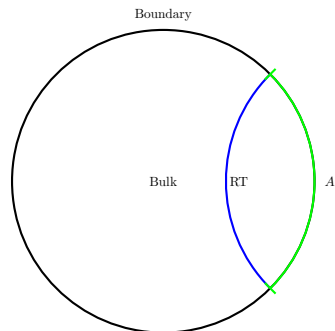


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By considering all possible subregions and their respective entanglement entropies, we may hope to reconstruct the dual bulk geometry from field theory data.

Holographic Complexity

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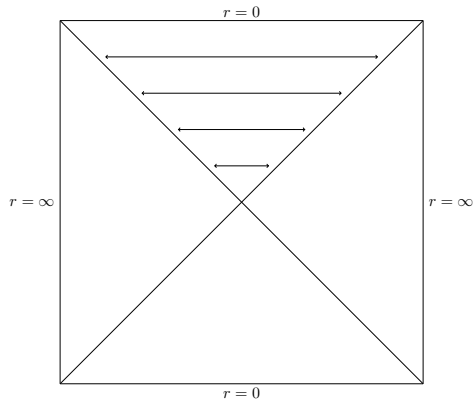


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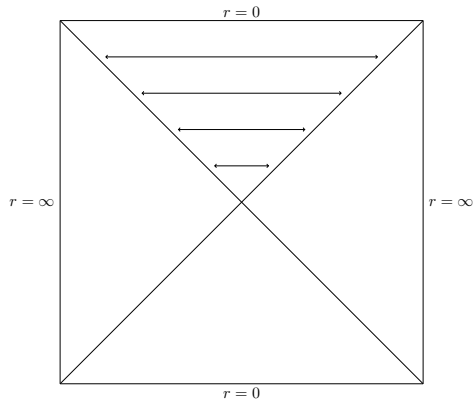


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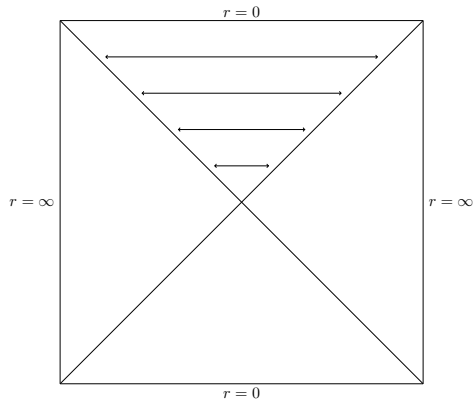


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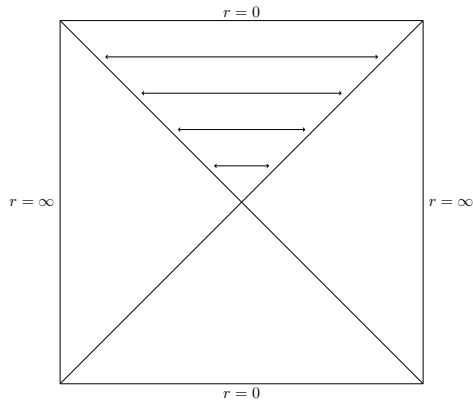


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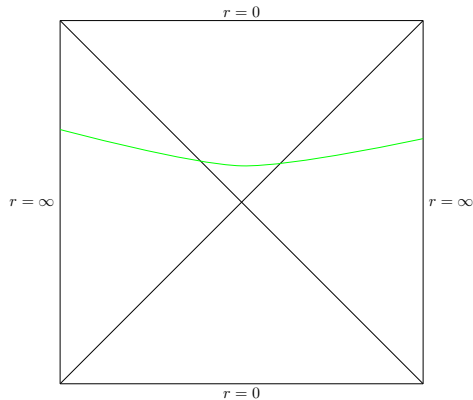


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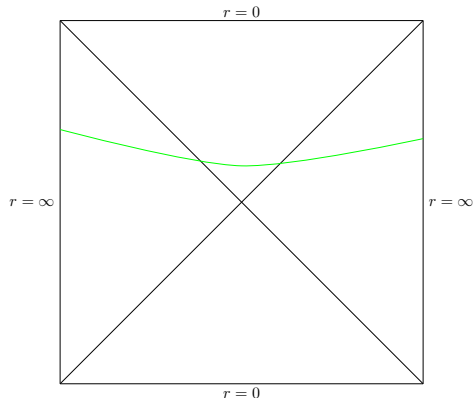


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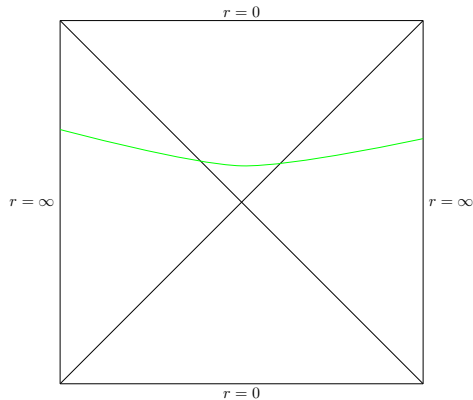


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- Later, 'complexity = action' (CA) was proposed as an alternative to CV, which states that complexity is dual to the action of the WDW patch

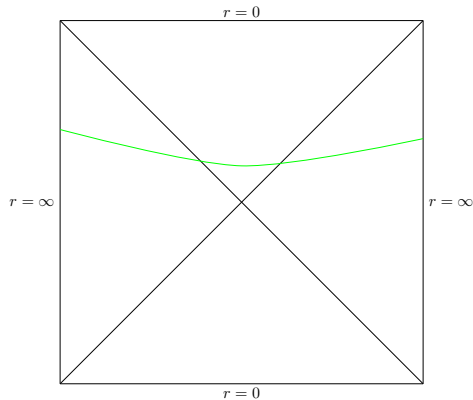


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- Complexity of a state (relative to a reference):

$$\mathcal{C}(|\psi\rangle) = \min_{U: U|r\rangle = |\psi\rangle} \mathcal{C}(U) \quad (3)$$

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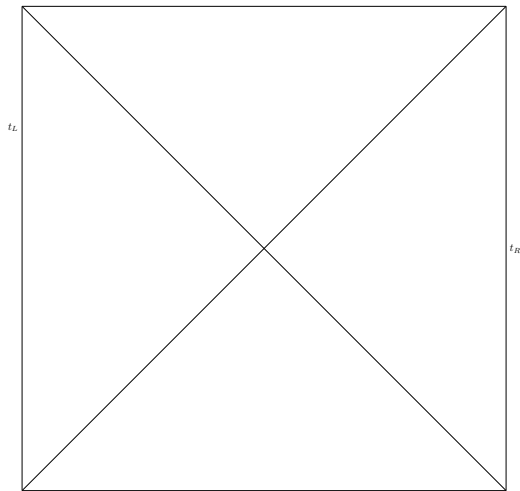


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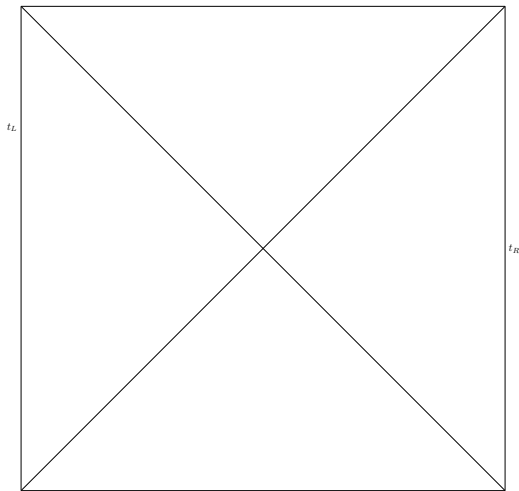


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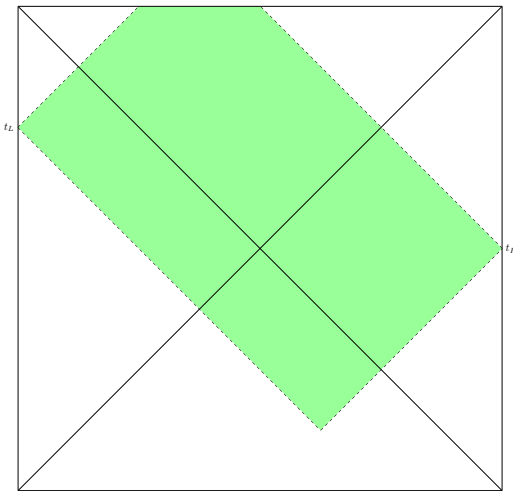


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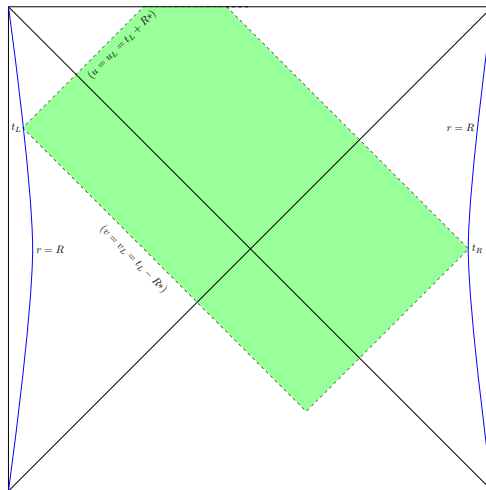


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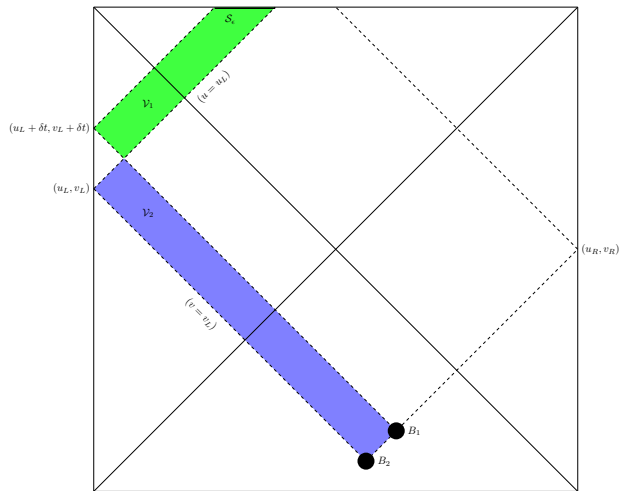


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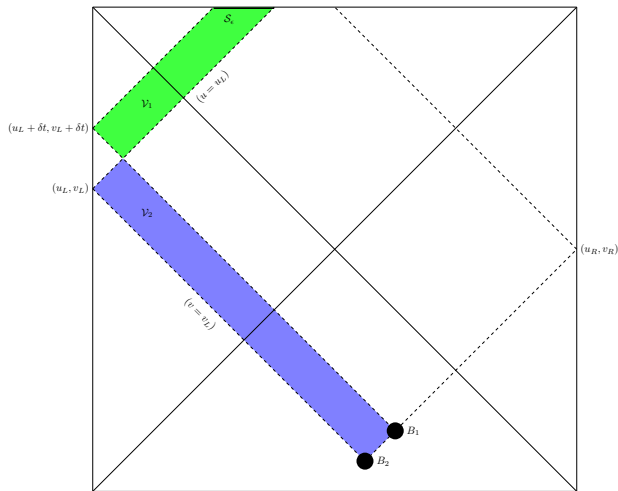


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- This difference of actions decomposes to the difference of two bulk pieces, a piece from the spacelike boundary of a near singularity cutoff, and two codimension two contributions from the past corners of the patches.

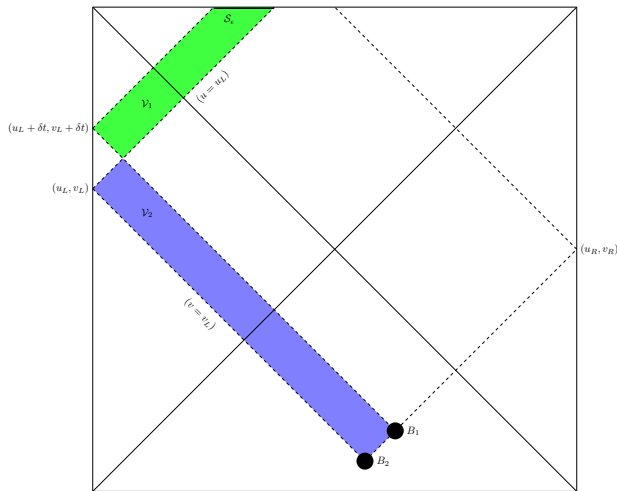


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- These conjectures come in two versions, 'complexity = action' and 'complexity = volume'.
- While there are qualitative arguments in favor of these conjectures, we do not have any solid demonstration that they are true.

Holographic Complexity and Extended Thermodynamics

Based on 1610.02038
Work with Willy Fischler and Phuc Nguyen

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- However, one can think of the cosmological constant as giving rise to a pressure according to

$$P = -\frac{\Lambda}{8\pi G} \quad (9)$$

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- By varying the cosmological constant Λ , and thus the pressure P , one get's an extended first law of the form

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- Instead, we ultimately attempted to interpret V_{th} through its relationship to holographic complexity

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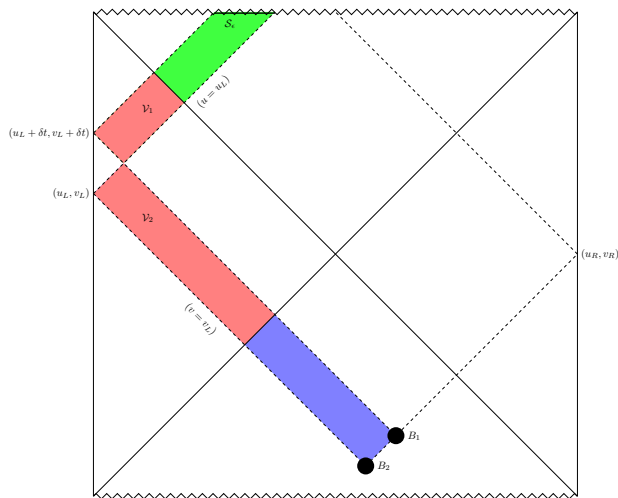


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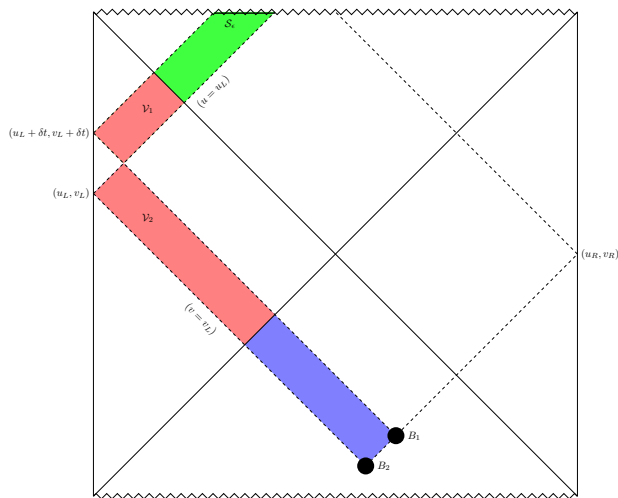


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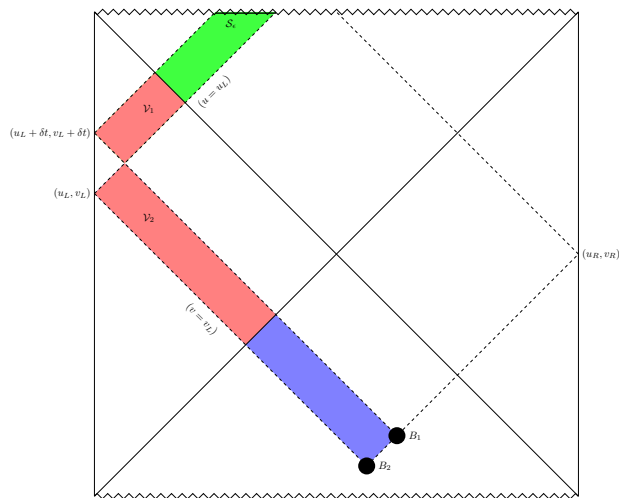


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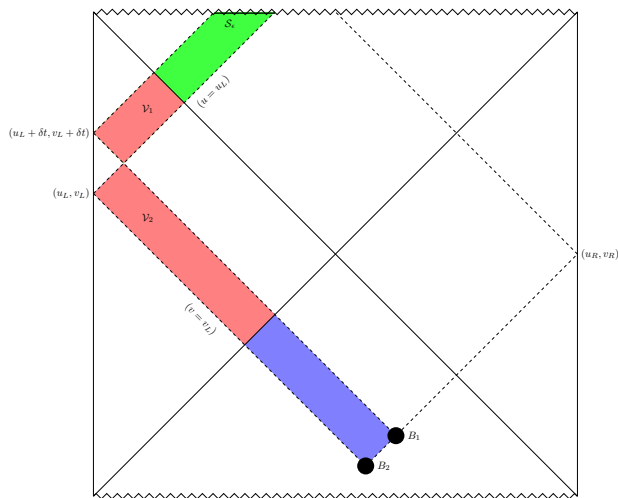


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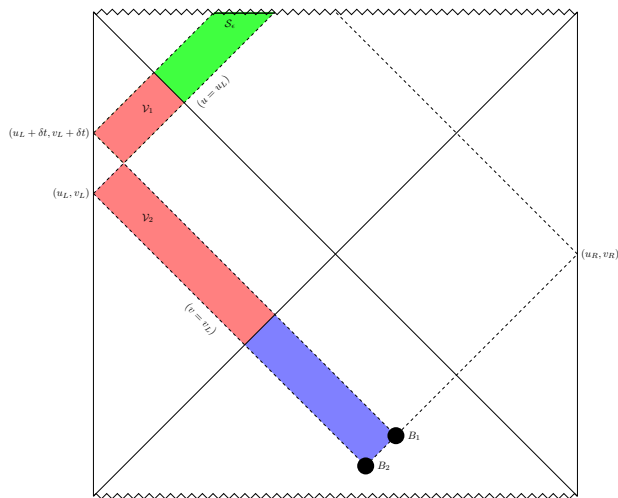


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- Finally, the bulk term in green gives (in the late time limit) a contribution of $-\frac{r_+^3}{2GL^2}\delta t = \boxed{PV_{th}}\delta t$!!

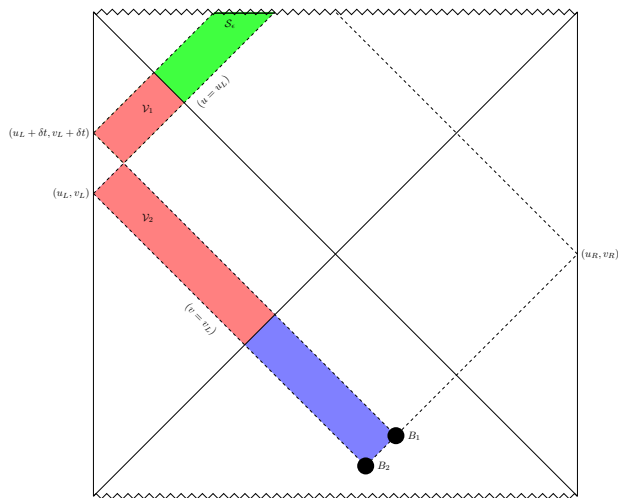


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CA for AdS-Ker

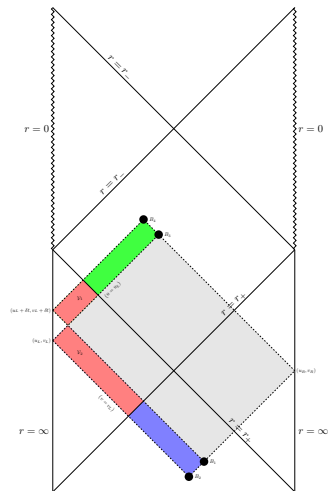


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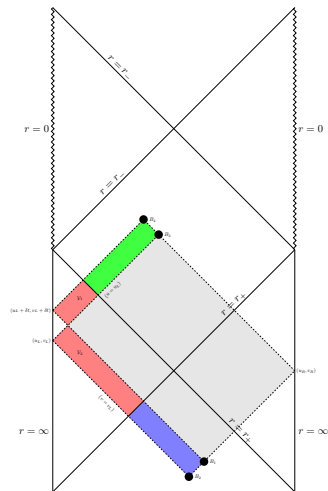


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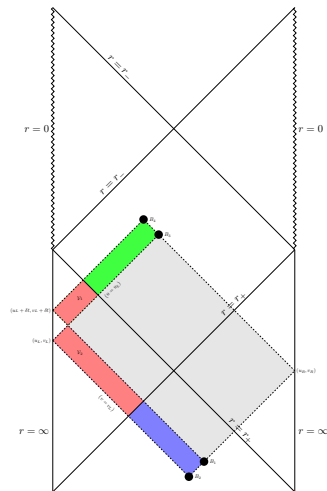


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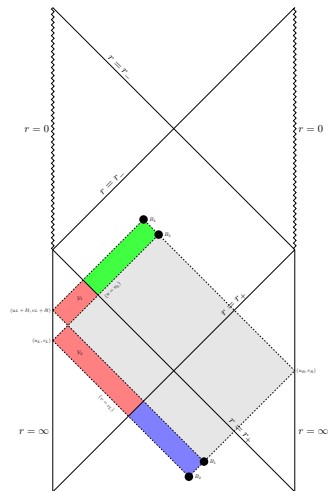


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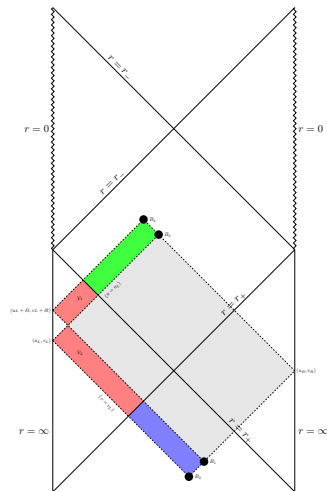


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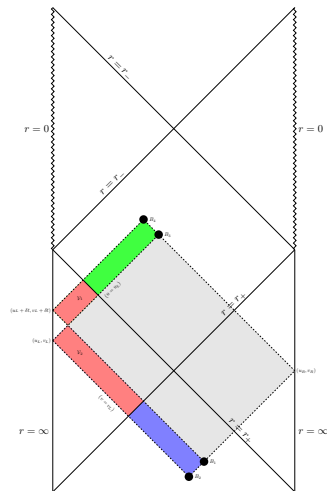


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Complexity = Spacetime Volume

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- We termed this 'complexity = spacetime volume' or CV2.0

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- If we take Δt to be the smallest time interval such that these states are orthogonal, and take the corresponding N to be 1, we get the Lloyd bound [5]:

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- Neither CA nor CV2.0 always obeys the Lloyd bound, though CV2.0 seems to obey it more often.
- Under either of CA or CV2.0, there seems to be a connection between extended thermodynamics and the rate of increase of complexity

Holographic Subregion Complexity and Purification Complexity

Based on 1811.10650

Work with Elena Cáceres, Stefan Eccles, and Willy Fischler

Subregion complexity

- The bulk side of the various holographic complexity proposals (CA, CV, CV2.0) can all be easily extended to a submanifold of some geodesically complete space

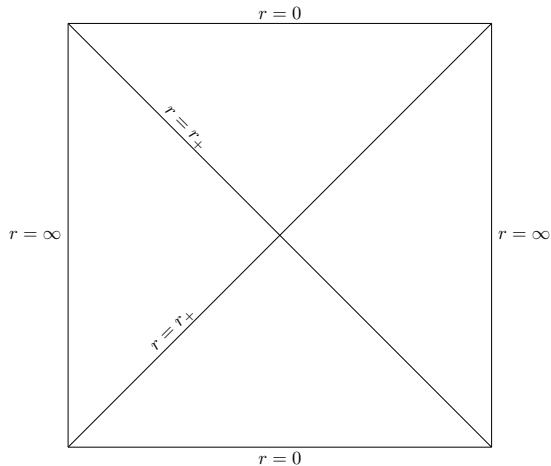


Figure: Two-sided black hole

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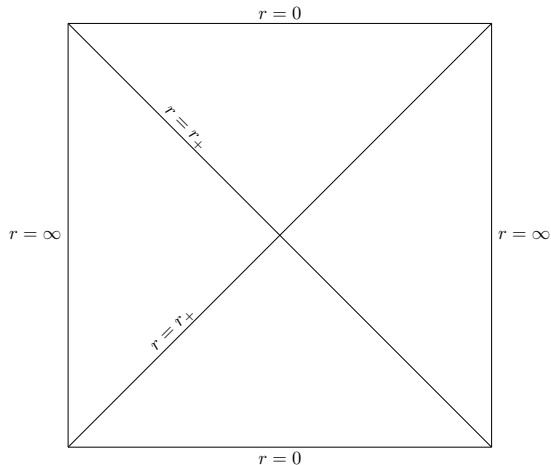


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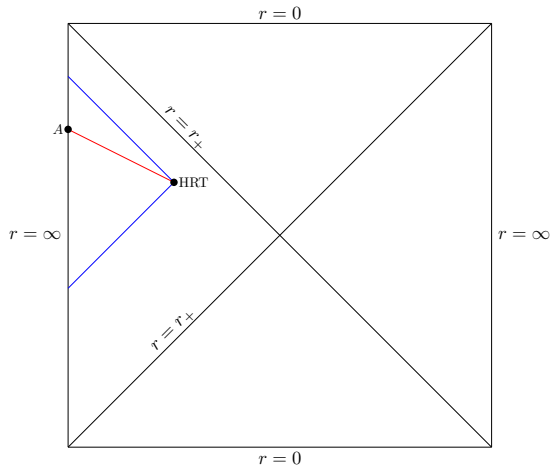


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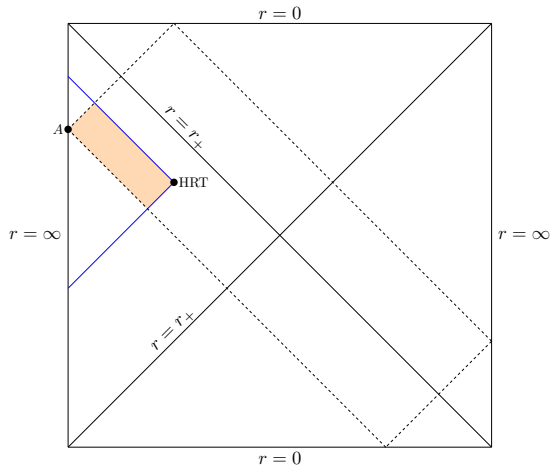


Figure: Subregion complexity: CA and CV2.0

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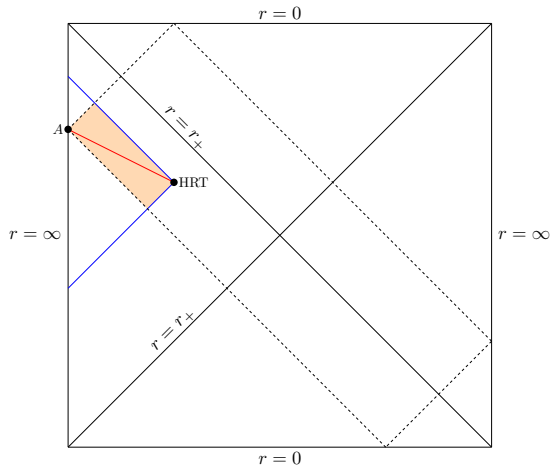


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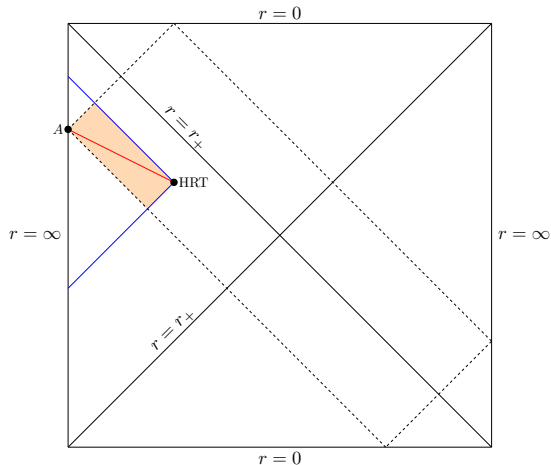


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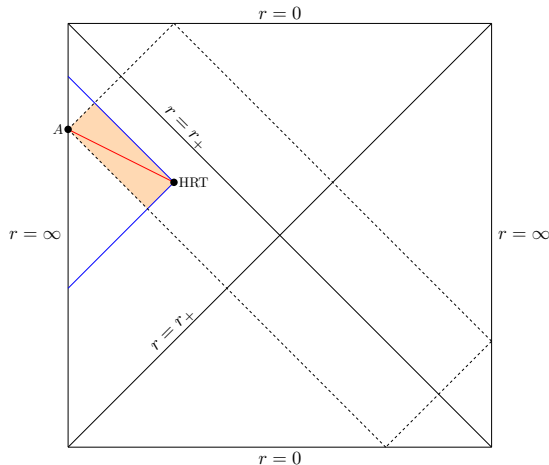


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- We can actually get a holographic bound on purification complexity by considering different holographic purifications.

Different holographic purifications

- Consider left side of 2-sided BTZ BH (at time t_L)

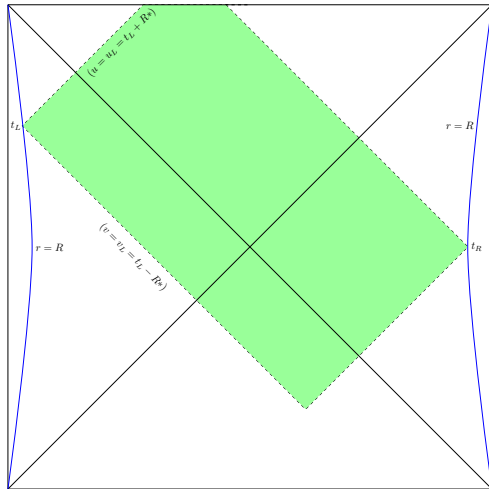


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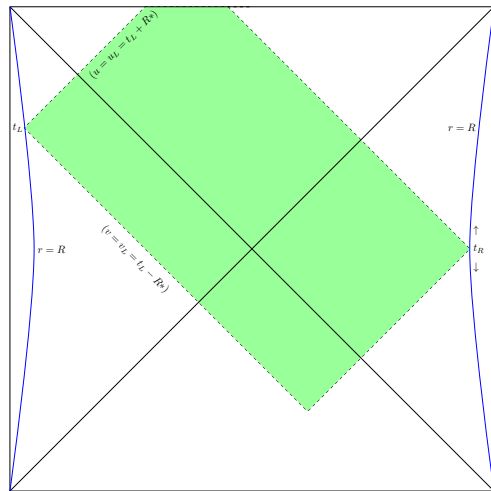


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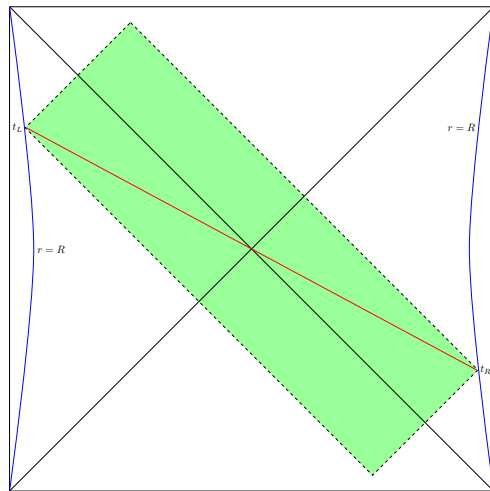


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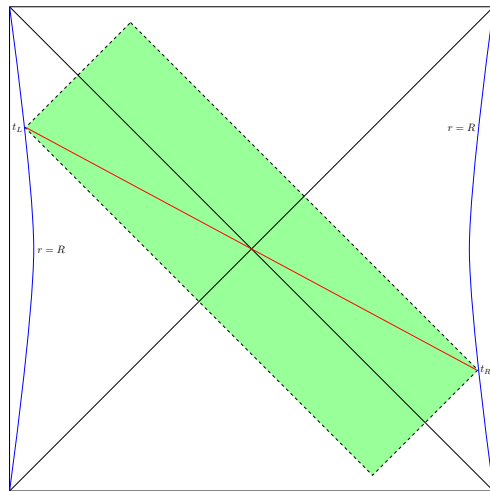


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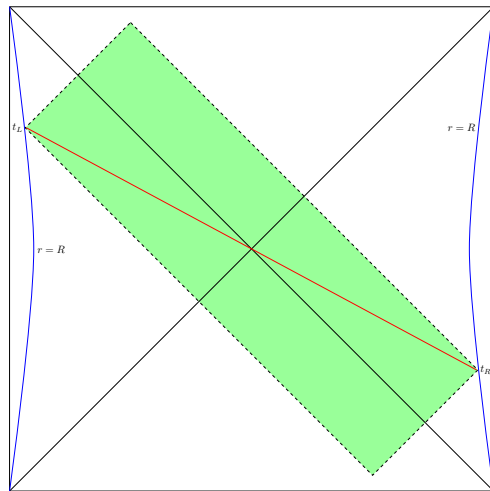


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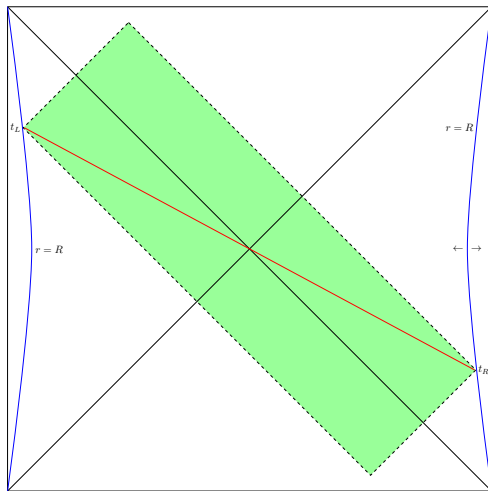


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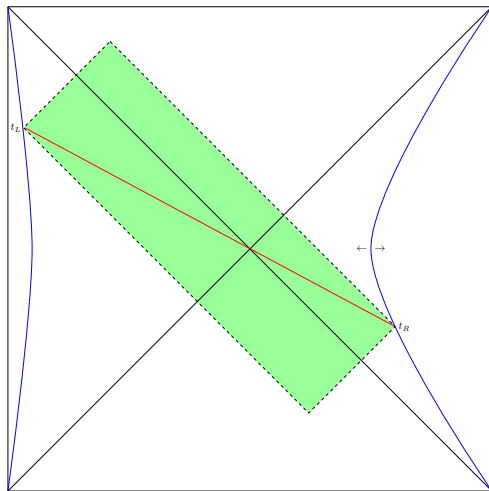


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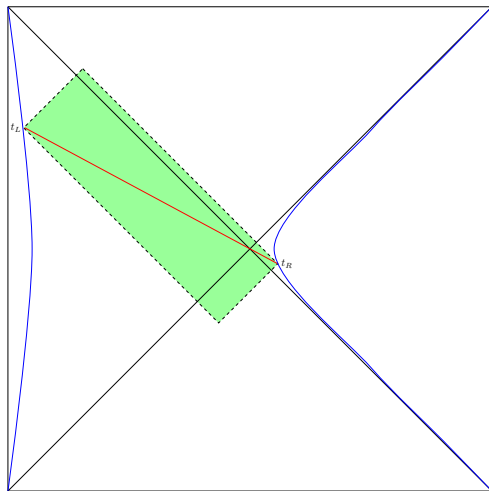


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- If this makes sense, the optimum is when cutoff hugs the horizon (For CV and CV2.0, CA less clear).

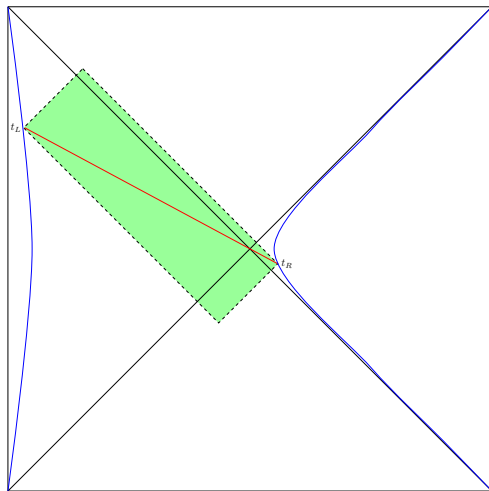


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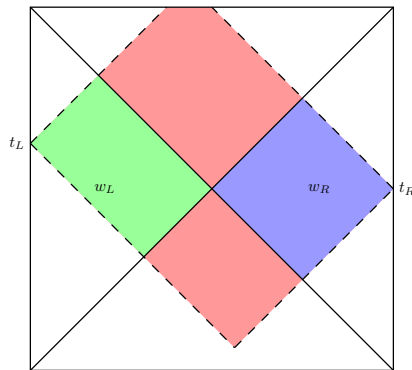


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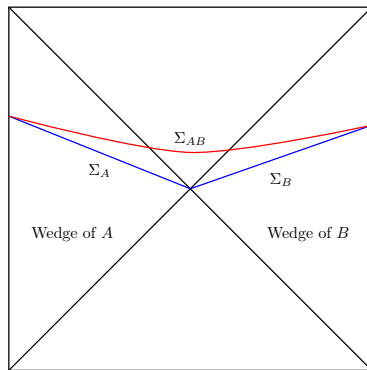


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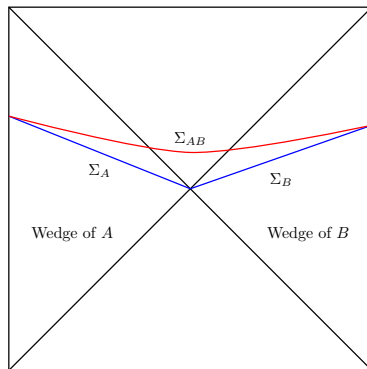


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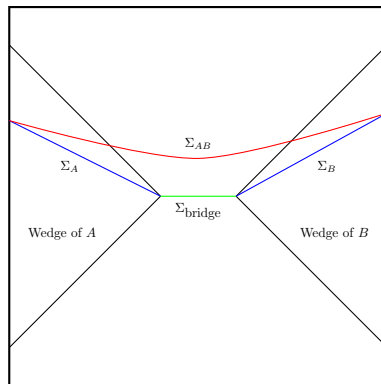


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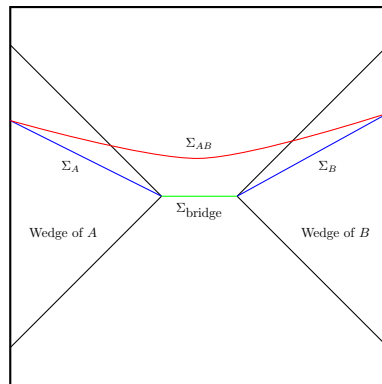


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- However, one might also argue that the parameter choices involved are for some reason not allowed

Does subregion complexity = purification complexity?

- CV and CV2.0: Pretty clear, if changing cutoff is okay.

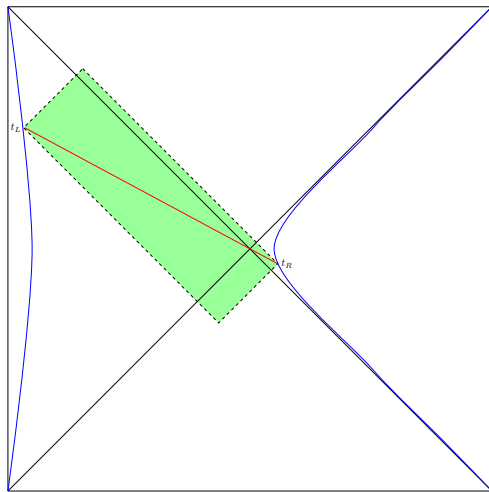


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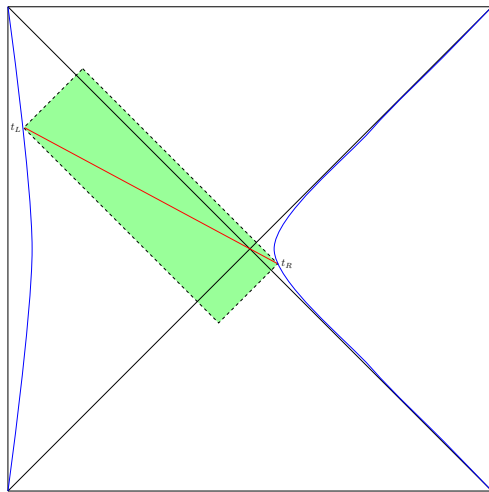


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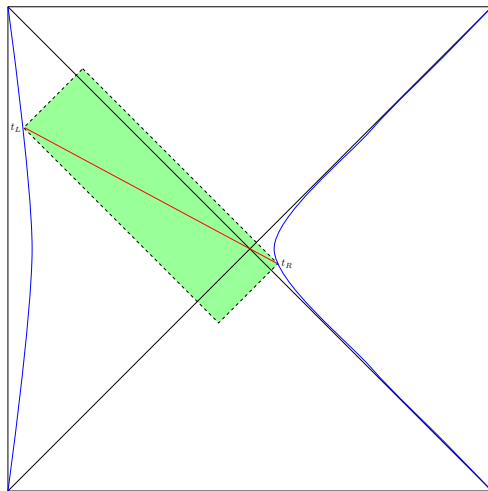


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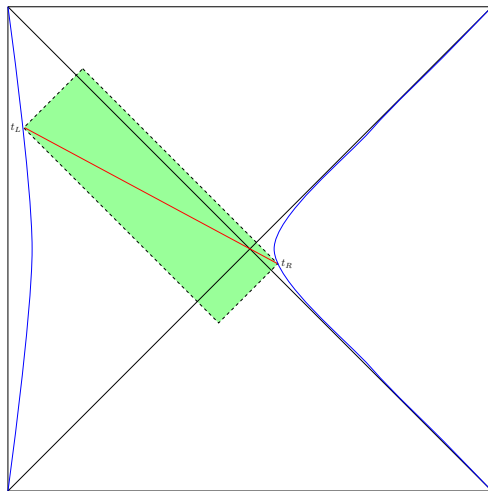


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Entanglement of Purification

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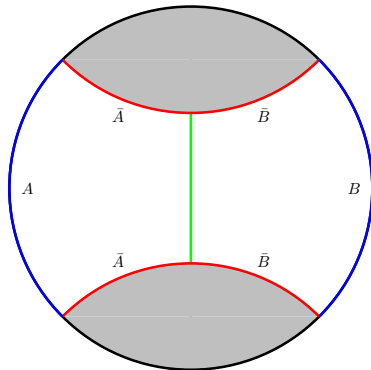


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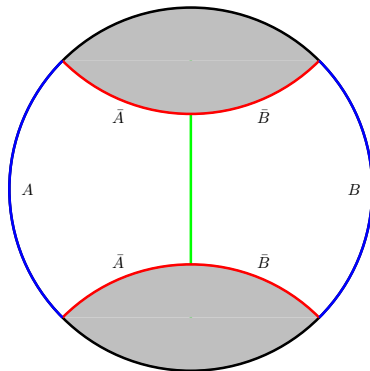


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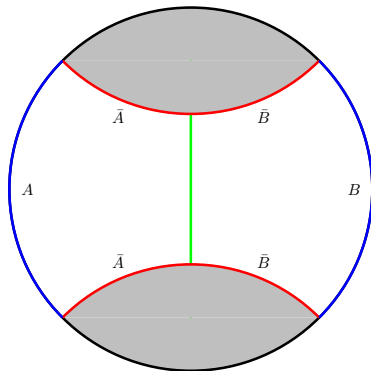


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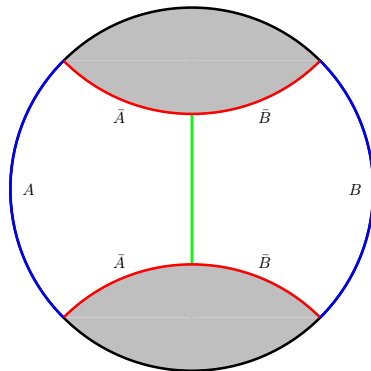


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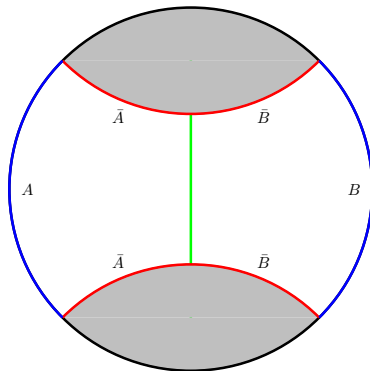


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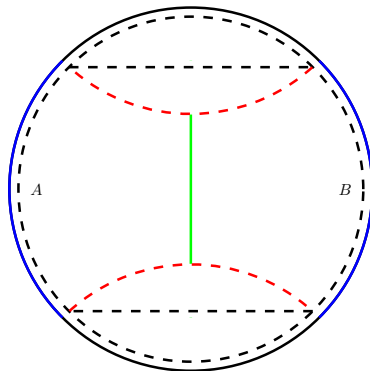


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- If we are allowed to change cutoff (for complimentary region), it is clear subregion complexity and purification complexity coincide for CV and CV2.0.

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References



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Behind the BH Horizon

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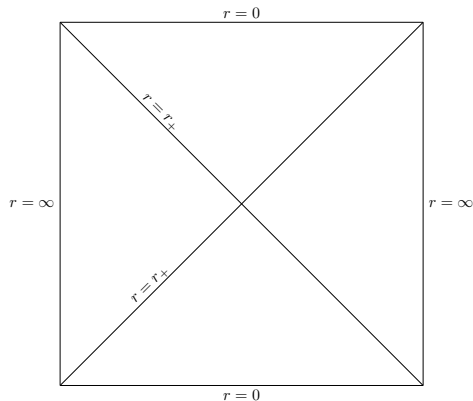


Figure: Two-sided black hole

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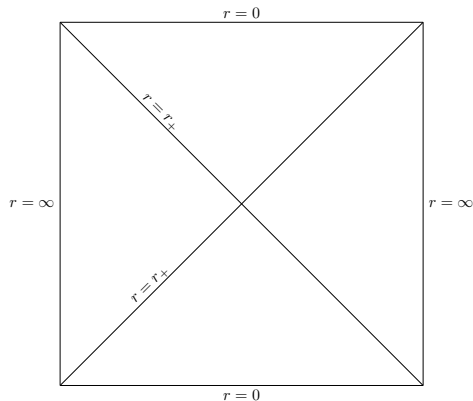


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- This geometry has two boundaries, with a CFT on each boundary. The Hilbert space is given by $\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R$

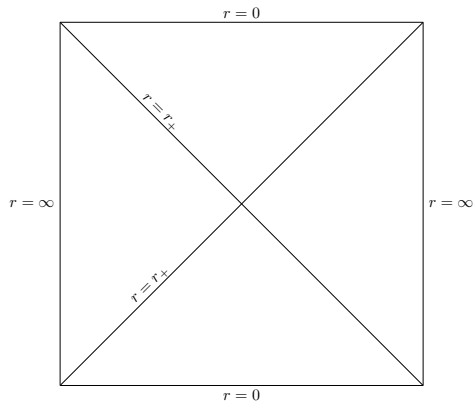


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$$\psi = \sum_n e^{-\beta E_n/2} |E_n\rangle_L \otimes |E_n\rangle_R$$

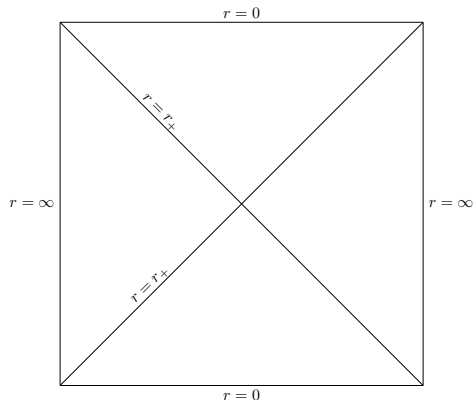


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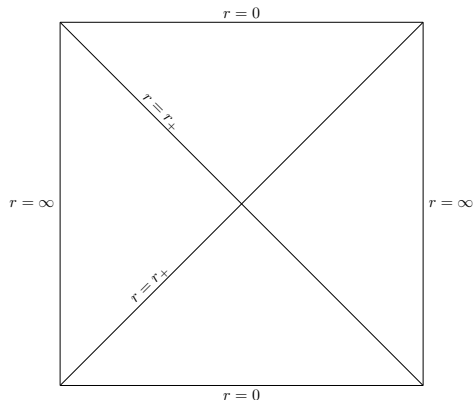


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- What about the geometry behind the horizon?

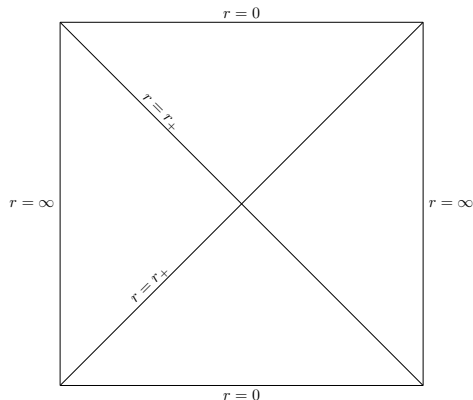


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 - ▶ For small W , generically get linear growth for $t > t_{\text{sat}}$
- This type of operator is dual to shockwave insertions in AdS
- CA and CV exhibit the expected behavior when applied to these shockwave geometries

Complexity = Action

Complexity = volume has a few unpleasant features.

- For example, in order to reproduce the correct boundary behavior, the volume must be multiplied by a non-universal length scale

One might seek an alternative proposal, which still captures something about the geometry behind the horizon, but which does not have these features. Such an alternative has been proposed by Susskind et al. [2, 3], and it goes by 'complexity = action.'

- According to complexity = action, the complexity is dual to the action of a so-called 'Wheeler-DeWitt' (WDW) patch.
- Because this is an action, it can be nondimensionalized with some multiple of $\hbar$.
- A universal choice for this coefficient is consistent with the expected large temperature behavior.
- Also exhibits linear growth at late time etc.